

Evaluation of Nasal Prongs for Estimating Nasal Flow

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Nasal prongs (NP) connected to a pressure transducer have been suggested as a useful alternative for measuring nasal flow in sleep apnea/hypopnea patients. However, flow measured with NP is expected to be nonlinear. The aim of the present study was to analyze and correct the nonlinearity of nasal flow measurements with NP ($\dot{V}_{NP}$). Nasal flow was simultaneously measured with a pneumotachograph (PNT; $\dot{V}$) and (NP; $\dot{V}_{NP}$) in six healthy subjects during 60 s of breathing at different tidal volumes. Nonlinearity of $\dot{V}_{NP}$ was assessed by fitting separately, for inspiration and expiration, a Rorher-model equation ($\dot{V}_{NP} \times K_1 \times \dot{V} + K_2 \times \dot{V}^2$). In addition, we fitted the data to a simpler nonlinear quadratic model ($P = K \times \dot{V}^2$). The main findings were: (1) an excellent fit of the Rorher equation to measured data in all cases; (2) although differences in the Rorher equation coefficients between inspiration and expiration were observed, they were not statistically significant; (3) a substantial intersubject variability was found; and finally, (4) the square root of $\dot{V}_{NP}$ acceptably fitted the nasal flow data measured by PNT ($\dot{V}$) in most cases. We conclude that in order to quantitatively assess nasal flow with NP, data should be corrected for their nonlinear pressure-flow relationships and, that the square root of the flow signal measured with NP is the simplest method of correcting for the observed nonlinearity. **Montserrat JM, Farré R, Ballester E, Felez MA, Pastó M, Navajas D. Evaluation of nasal prongs for estimating nasal flow.**

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Conventional sleep studies have extensively used thermally sensitive devices — thermistors — to monitor nasal and mouth airflow (1). Depending on the qualitative changes in the thermistor signal together with the thoracoabdominal motion, apneas and hypopneas are usually defined. Apnea definition is a well-established issue (cessation of airflow). By contrast, a debate exists on how to define hypopnea (2). A 50% reduction of airflow signal or thoracoabdominal motion followed by a SAO_2 dip and/or an arousal is the common accepted definition. However, as the recent description of the upper airway resistance syndrome has pointed out (3), thermistors cannot detect the full range of flow-related respiratory events that occur throughout the night. In addition, because of their long time-constant response (4), thermistors do not allow the analysis of the shape of the inspiratory flow, which we and other authors have observed to be a reliable method for detecting increases in upper airway resistance (5, 6). In addition, we have observed that thermistors underestimate changes in nasal flow (7). As an alternative to thermistors, it has recently been suggested that nasal prongs (NP) connected to a pressure transducer could be a useful way to measure nasal flow (8). Because of their short time constant, the main advantage of NP would be to allow the analysis of the time morphology of flow. Moreover,

NP are very simple to use for sleep-diagnostic studies as compared with pneumotachographs (PNTs). However, suspected nonlinearity could modify the estimate of nasal flow through NP as a consequence of the flow-dynamics phenomena involved. In this regard, the present study was designed to analyze the capability of NP to assess nasal flow.

METHODS

Patient Population

Six subjects (five males and one female), aged 28 to 45 yr and in good health, without a history of sleep disturbances and with a normal spirometry, were selected for the study. They were staff members or fellows in the University of Barcelona who volunteered for the study with informed consent. The protocol of the study was approved by the Ethics Committee of Clínic Hospital.

Experimental Protocol

Subjects in the sitting position were asked to breathe exclusively through the nose, which was connected concurrently to a facial mask and NP (Figure 1). The facial mask was accurately adjusted to the face of the subject, taking special care to avoid leaks. To measure flow, the facial mask was connected to a PNT (Fleisch, Lausanne, Switzerland) connected to a differential pressure transducer (MP45, ± 2 cm H_2O ; Validyne Co., Northridge, CA). The NP had a length of 40 cm and were connected to a differential pressure transducer (MP45, ± 5 cm H_2O ; Validyne Co.). The flow measured with the NP was estimated as the pressure referred to the mask pressure (Figure 1).

Nasal flow was recorded simultaneously with the PNT ($\dot{V}$) and with the NP ($\dot{V}_{NP}$) during 60 s. This record consisted of 20 s of normal breathing followed by 20 s of increased tidal breathing and 20 s of decreased tidal breathing. An example of the recorded flow signals is shown in Figure 2. For the sake of comparison, $\dot{V}_{NP}$ in Figure 2 was scaled so that its amplitude during the first breathing cycle coincided with $\dot{V}$.

To assess the variability, measurements were carried out in two inde-

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Figure 1. Experimental setup. Subjects breathed through a facial mask that was accurately fitted on the face. Flow was measured through a pneumotachograph and by nasal prongs connected to a pressure transducer. Flow measured with the nasal prongs was estimated as the pressure referred to mask pressure.

pendent runs. Between both runs, the mask and NP were completely removed from the subject's face and the subject was allowed to breathe spontaneously for 5 min before the next recording. NP nonlinearity was assessed by fitting a Rorher-model equation ($\dot{V}_{NP} = K_1 \times \dot{V} + K_2 \times \dot{V}^2$) separately for inspiration and expiration. We also used a simplified quadratic model ($\dot{V}_{NP} = K \times \dot{V}^2$) to assess the relationship between $\dot{V}_{NP}$ and $\dot{V}$. As will be discussed subsequently, such a model would facilitate and simplify clinical applications of NP.

Statistical Analysis

Statistical analysis was performed with SPSS software (Statistical Package for Social Sciences; SPSS Inc., Chicago, IL). Data were expressed as mean $\pm$ SD. Comparison between K_1 and K_2 at the two different runs was made using Student's *t* test. The variability of K_1 and K_2 at the two different runs was assessed by comparing, the sum of $K_1 + K_2$ (which corresponds to the NP signal at 1 L/s), separately for inspiration and expiration, between the two different runs. Statistical significance was accepted at $p < 0.05$.

RESULTS

As shown in Figure 3, we found in all cases an excellent fit of the Rorher equation to the measured data, both in inspiration

and expiration. Table 1 shows K_1 and K_2 during inspiration and expiration for the two independent runs. Although the differences were not statistically significant, the Rorher model and coefficients were different in the two runs. Table 2 shows the percent difference of the sum of K_1 and K_2 between the two different runs, both for inspiration and expiration. Considerable differences were found, in some cases of more than 100%.

To illustrate the adequacy of analyzing NP data with the simplified quadratic model, we plotted the square root of $\dot{V}_{NP}$ versus $\dot{V}$ obtained with PNT (Figure 4). Compared with Figure 3, these plots are almost lineal, especially during the inspiratory component, which is the interesting breath phase to look at in patients with sleep apnea/hypopnea syndrome.

The effect of computing the square root of $\dot{V}_{NP}$ is shown in the example in Figure 2, in which its amplitude was scaled to achieve the same amplitude as $\dot{V}$ in the first breathing cycle. This plot shows that the square root of the flow signal obtained with NP was almost coincident with the actual flow signal obtained with the PNT.

In view of the potential application of NP for assessing flow during sleep studies, and specifically for the detection of hypopneas, we considered the situation of a typical inspiratory flow amplitude of $\dot{V} = 0.5$ L/s and a reduction corresponding to a hypopnea of $\dot{V} = 0.25$ L/s. When the NP pressure corresponding to these flow values is computed from the Rorher equation by using K_1 and K_2 for the first run in Table 1, the NP signal decreases from 1.57 ± 1.03 cm H₂O to 0.49 ± 0.31 cm H₂O. Therefore, due to NP nonlinearity, actual flow reduction is considerably overestimated: $69 \pm 3\%$ instead of 50%. By contrast, taking the square root of these NP values results in a decrease of $44 \pm 3\%$, which slightly overestimates the actual flow reduction (50%). Consequently, the magnitude of the bias error in the NP signal is reduced to one third (from 19% to 6%). Therefore, the square-root correction of the NP signal allows a reasonable estimate of true nasal flow. The same results were obtained when we analyzed the data from the second independent run (Table 1).

DISCUSSION

Our study was designed to evaluate the performance of NP for assessing nasal flow and to model the NP response in order to correct for possible nonlinearities. We found a nonlinear relationship between the NP signal and the PNT flow, and an outstanding fit of the Rorher equation to the measured data. In addition, we observed differences in the Rorher equation parameters between inspiration and expiration and between the two separate runs. When the flow signal obtained with NP was corrected by its square root, a quasilinear relationship was observed between the squared NP signal and flow measured with the PNT.

Thermistors are the conventional systems to monitor nasal and mouth airflow in sleep studies (1). However, it is well known that the signal obtained with the thermistors usually employed in sleep studies have a poor time response. The result is a delay in the apparent timing of respiration in relation to the intrathoracic pressure changes (measured by esophageal pressure), thus limiting the accuracy of the procedure. This delay has been observed to vary from 120 to 720 ms, with a mean of 370 ms, and to occur at different breathing frequencies (4). Consequently, thermistors, because of their prolonged time constant (near 1 s), do not allow analysis of the morphology of inspiratory flow, which we and others have proved to be a reliable method for estimating the lowest esophageal pressure during continuous positive airway pressure (CPAP) titration (5, 6).

Recently, NP have been suggested as a useful alternative to

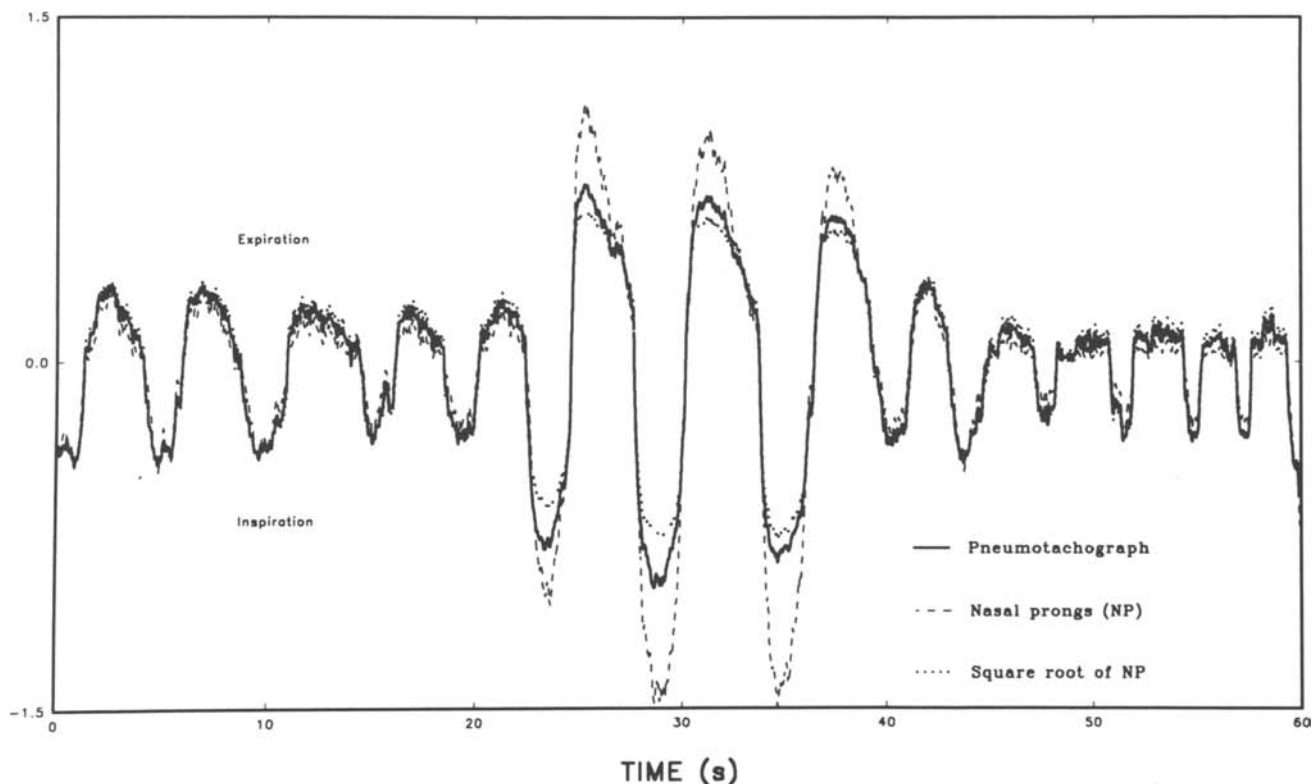


Figure 2. Example of nasal flow recorded simultaneously with a pneumotachograph (solid line) and nasal prongs (dashed line). The dotted line represents the square root of the nasal prongs signal. For comparison, the amplitude of the nasal prongs signal and of its square root were scaled to achieve the same amplitude as with the pneumotachograph in the first breathing cycle. Note that the signal from the nasal prongs overestimates the pneumotachograph signal, and that after the square root correction, the nasal prongs signal was almost coincident with the pneumotachograph signal.

thermistors for measuring nasal flow (8). Nevertheless, NP may present some problems in accurately assessing nasal flow. Indeed, the capability of NP to detect flow is based on the sensing of pressure changes associated with turbulence at the nostrils. Therefore, the effective resistance appearing between the NP lumen and the atmosphere acts in a similar manner to the resistance of a PNT. However, since this resistance is mainly due to turbulence, the NP signal would be expected to exhibit a nonlinear relationship with actual flow at the nose. Another potential concern with NP is their inability to detect mouth breathing during sleep.

Because of the excellent dynamic response of NP, flow measured with them should permit the analysis of both flow and its morphology. Moreover, an additional advantage over the indirect thermistor output signal is the detection of snoring, which is an airflow-related phenomenon that produces a high-frequency vibration that is of considerable interest in sleep studies (9). Partial airflow limitation (inspiratory flow-contour flattening) and snoring-related vibrations are best observed from the flow morphology.

In the present study we have demonstrated that nasal flow estimated from NP has a nonlinear behavior and follows a Rorher model. We modeled the nonlinearity of the flow obtained with NP with a quadratic Rorher equation, since this kind of nonlinearity is the kind most commonly found in respiratory system airflow studies (10, 11). We observed that the Rorher coefficients were different during inspiration and expiration. Therefore, the proper procedure to correct for such a nonlinear relationship would be to calibrate and then apply the Rorher coefficients both

to inspiration and expiration for each patient. Moreover, bearing in mind the differences between the Rorher coefficients in the two different runs, it is reasonable to expect that during the night, as a consequence of NP movement, the Rorher coefficients would change and consequently a new calibration would be required. This would obviously make difficult or even impossible the use of NP during routine sleep studies.

To simplify the correction of nonlinearity, we considered the possibility of applying a simplified quadratic model that considers only the turbulent component of the NP signal ($\dot{V}_{NP} = K \times \dot{V}^2$). When analyzing this approach, we found that the square root of the NP signal provided a reasonable linearized flow assessment. Interestingly, this linearization was better (Figure 4) for the inspiratory phase, in which the quadratic coefficient K_2 plays a major role as compared with that of K_1 (Table 1). As shown in the RESULTS section, this linearization would reduce the error in the detection of hypopneas from 19% to 6%.

In conclusion, our results suggest that use of the square root of the NP signal is a simple procedure for accurately assessing nasal flow. Therefore, despite the possible shortcoming of failure to detect mouth breathing, NP have a potential application during sleep studies because of the following advantages: (1) NP are easily tolerated; (2) the square root of the NP signal can be easily computed digitally or with a simple analogic circuit when an analogic recorder is used; (3) as opposed to thermistors, the excellent time response of NP allows the precise detection of each different component of or event within the breathing cycle; and (4) no calibration is required to detect relative changes in nasal flow.

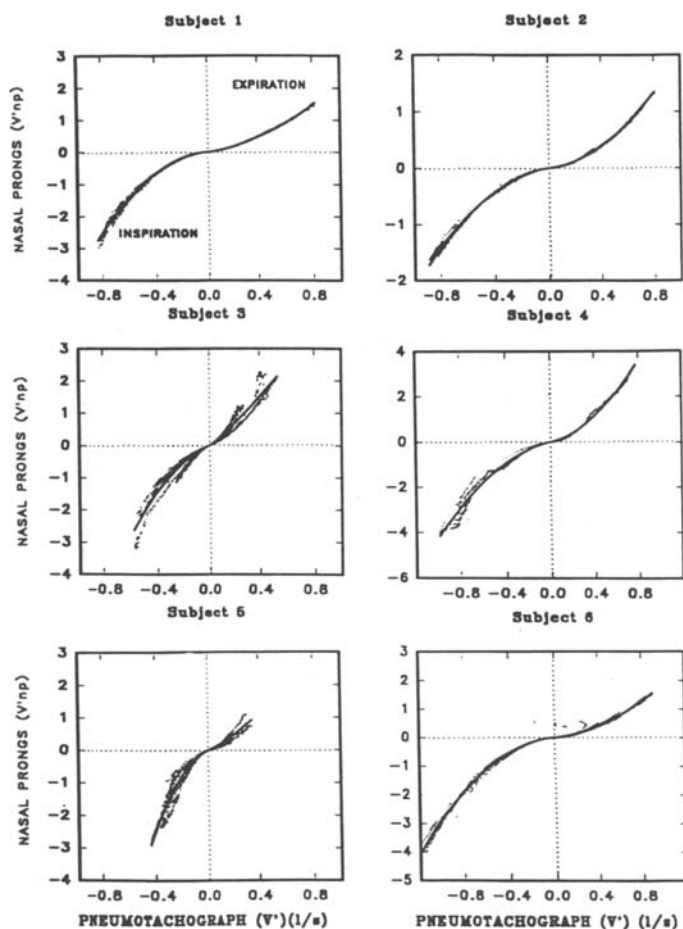


Figure 3. The nonlinearity of the flow obtained with nasal prongs was assessed by fitting, separately for inspiration and expiration, a Rorher model (solid lines) to the measured data (dotted lines).

TABLE 1
RORHER EQUATION PARAMETERS DURING INSPIRATION
AND EXPIRATION FOR THE TWO DIFFERENT RUNS

Subjects	Inspiratory K_1^*	Inspiratory $K_2^\dagger$	Expiratory K_1	Expiratory K_2
First run				
1	0.09	3.85	0.49	1.71
2	0.32	1.81	0.31	1.70
3	1.53	5.08	2.86	2.26
4	1.02	3.28	1.25	4.05
5	1.29	12.0	1.56	3.18
6	0.32	2.56	0.42	1.47
Mean $\pm$ SD	0.76 ± 0.6	4.76 ± 3.7	1.14 ± 1.0	2.39 ± 1.0
Second run				
1	0.15	4.96	0.51	2.09
2	0.92	2.65	1.03	2.31
3	0.88	3.93	1.54	3.93
4	1.75	8.49	2.14	4.39
5	1.97	10.7	2.12	4.72
6	0.22	2.30	0.52	0.76
Mean $\pm$ SD	0.98 ± 0.7	5.50 ± 3.4	1.31 ± 0.74	3.03 ± 1.55

* In cm H₂O/(L/s).

† In cm H₂O/(L/s)².

TABLE 2
PERCENT DIFFERENCES OF THE SUM OF RORHER
COEFFICIENTS BETWEEN THE TWO DIFFERENT RUNS
DURING BOTH INSPIRATION AND EXPIRATION

Subjects	First Run $K_1 + K_2^*$	Second Run $K_1 + K_2$	% Change
1	3.94	5.11	29%
2	2.13	3.57	67%
3	6.61	4.81	27%
4	4.30	10.2	137%
5	13.3	12.7	4.5%
6	2.88	2.52	12%
Mean $\pm$ SD	5.5 ± 3.7	6.5 ± 3.6	46 ± 45.2
Expiration			
1	2.20	2.60	18%
2	2.00	3.34	67%
3	5.12	5.47	6.8%
4	5.30	6.53	23%
5	4.74	6.84	44%
6	1.89	1.28	32%
Mean $\pm$ SD	3.5 ± 1	4.3 ± 2	32 ± 19

* $K_1 + K_2$: NP signal at a flow of 1 L/s.

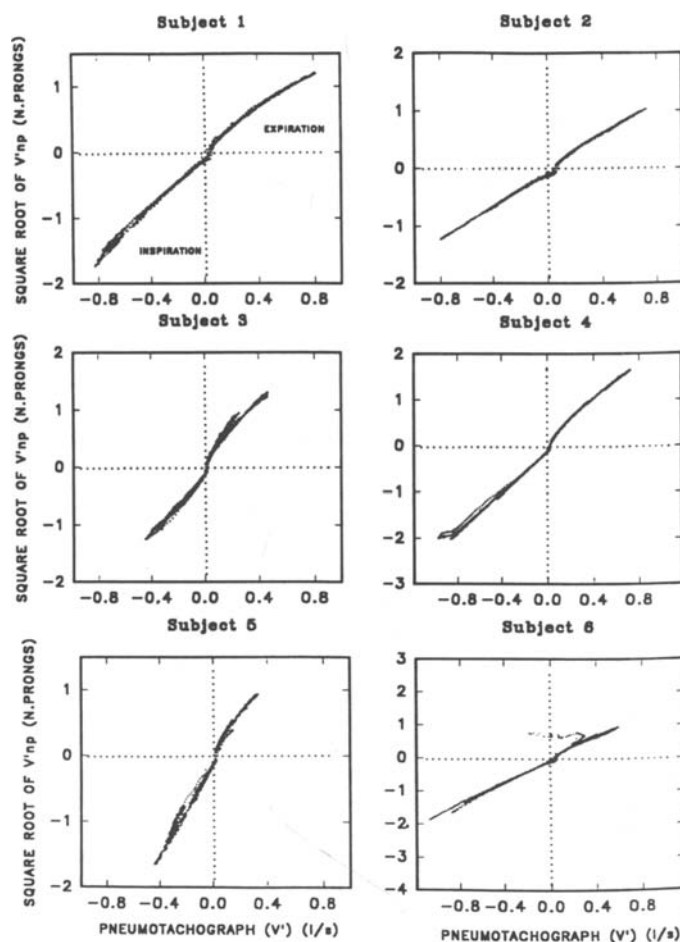


Figure 4. Relationship between the square root of the flow obtained with nasal prongs (V_{np}) and the flow obtained with a pneumotachograph ($\dot{V}$).

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